

What the loss hid

The task coordinate was written off on the strength of a denoising loss that barely moves. In generated samples it decides the outcome every time.

cifar.ds3 @ 50k · validation · DDIM $\eta = 0$, 50 steps · generation test 24×128 · strategy table 60×64

100 %

of generated sets carry the **right transformation** given the episode's own coordinate — against 33 % chance

1.7 %

is all the **same coordinate** is worth on the denoising loss the project has been reading

4.6×

closer to the target domain than reusing the source coordinate, at $K_T = 1$

The picture without any instrument

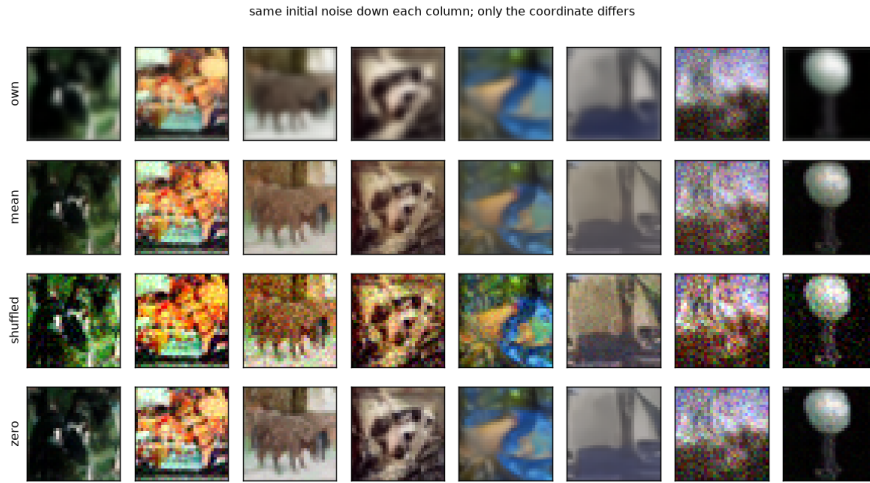


Figure 1: Each column starts from **one shared initial noise**, so composition is fixed down the column and only the coordinate changes. Columns 1–6 are blur tasks, 7–8 noise tasks. The **own** row follows the task; **mean** and **zero** ignore it; **shuffled** follows the *wrong* task.

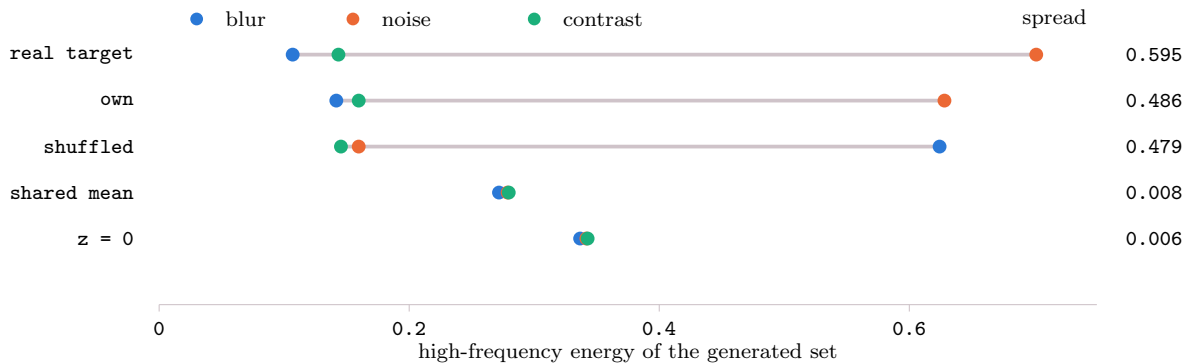


Figure 2: Blur suppresses high-frequency energy, added noise raises it. A coordinate that carries the task spreads its three values apart. **own** reproduces 82 % of the real spread; **mean** and **zero** collapse onto a point; **shuffled** spreads just as far, in the wrong direction — which is what rules out every explanation that does not run through the coordinate.

Deployment strategies, re-run in this metric

Distance from the real target domain, lower is better, over 60 episodes per K_T . The denoising loss puts all six rows on the same number.

strategy	$K_T = 1$	$K_T = 20$
no adaptation	1.618	1.637
reuse z_S	1.448	0.575
transport	0.316	0.346
transport + refine	0.314	0.357
target only	0.377	0.328
oracle	0.298	0.302

- **Transport beats reusing the source coordinate** by $+1.132 \pm 0.114$ at $K_T = 1$, falling to $+0.230 \pm 0.056$ at $K_T = 20$. Both intervals clear of zero.
- **Transport also beats encoding the target images, but only while they are scarce:** $+0.060 \pm 0.047$ at $K_T = 1$ — clear of zero — and -0.017 ± 0.044 at $K_T = 20$, where target-only has caught up. That crossing is the regime the method claims to own, and it is the first time it has been seen on images.
- **Refinement changes nothing measurable** ($+0.002 \pm 0.013$ and -0.012 ± 0.044). It costs 25 optimisation steps per deployment and buys no improvement in the samples.

A correction worth recording. An earlier pass with 12 episodes per K_T got both of those readings wrong, in opposite directions: it put transport level with target-only ($+0.027 \pm 0.096$) and made refinement look actively harmful (-0.085 ± 0.074 at $K_T = 20$). Neither survived five times the sample. At that size the interval on this comparison is wider than the effect, so both the null and the trend were over-read.

What this changes

- **The representation was never the problem.** The instrument that said so compresses the effect by roughly fifty-fold. The earlier “0 % task-specific” result reproduces exactly on the arm it was measured on — one transformation, 32 channels — and does not generalise to the arm where tasks differ in something denoising depends on.
- **The claim the method is built on now has evidence on images.** Source-informed transport beats both baselines at $K_T = 1$ and is overtaken as target evidence grows — the shape stage 1 produced analytically, reproduced where it counts.
- **Refinement can be switched off** in this setting. It is not harmful, but it buys nothing and costs 25 optimisation steps per deployment.
- **The centring change is no longer the indicated fix.** It was the remedy for a representation with no task-specific content, and that diagnosis has not survived.

What these numbers do not say

- Nothing clean about **semantic class**: the coordinate carries transformation identity ($+0.00133 \pm 0.00035$, 98.5 % of the effect) and almost no class identity ($+0.00002 \pm 0.00001$, 1.5 %).
- Generation reaches **82 %** of the way to the target domain, not all of it — and the oracle does no better, so the residual gap is the model’s reach rather than a mis-set coordinate.
- Everything here is the **three-transformation checkpoint**. The single-transformation arms show $\Delta_{\text{task}} \approx 0$ and have not been put through the generation test.

branch jinjian · scripts/posthoc_controls.py, gen_specificity.py, gen_strategies.py · raw output in handoff/Jinjian/results/